

Package ‘IFFuzzyStats’

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Type Package

Title Intuitionistic Fuzzy Statistics Toolkit

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Description A comprehensive toolkit for intuitionistic fuzzy sets (Atanassov, 1986): constructors and arithmetics for intuitionistic fuzzy values, Xu and Yager aggregation operators (IFWA, IFWG, IFOWA, IFOWG, IFHA, IFHG, Einstein), entropy and distance/similarity measures (Szmidt-Kacprzyk, Vlachos-Sergiadis, Ye, Hamming, Euclidean, Hausdorff and others), intuitionistic fuzzy c-means and k-means clustering, approximate reasoning / compositional inference, and multi-criteria decision helpers (IF-TOPSIS, score ranking). All computations use only base R with fully vectorised implementations.

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IFFuzzyStats-package *Intuitionistic Fuzzy Statistics Toolkit*

Description

A comprehensive toolkit for intuitionistic fuzzy sets (Atanassov, 1986), where each element carries membership μ , non-membership ν and hesitation $\pi = 1 - \mu - \nu$. Provides constructors and arithmetic, Xu/Yager aggregation operators, entropy and similarity measures, fuzzy clustering, compositional inference, and IF-TOPSIS.

References

Atanassov, K. (1986). Intuitionistic fuzzy sets. *Fuzzy Sets and Systems*, 20(1), 87-96. Xu, Z. (2007). Intuitionistic fuzzy aggregation operators. *IEEE Trans. Fuzzy Systems*, 15(6), 1179-1187. Szmidt, E. & Kacprzyk, J. (2000). Distances between intuitionistic fuzzy sets. *Fuzzy Sets and Systems*, 114(3), 505-518.

ifcmeans *Intuitionistic fuzzy c-means (IFCM) clustering*

Description

Clusters ifv data with intuitionistic fuzzy memberships. Distance between an observation a and centre v is the normalised Euclidean IF distance (Szmidt-Kacprzyk). An optional hesitation weight lambda inflates distances by $\exp(\lambda \cdot \bar{\pi})$ where $\bar{\pi}$ is the mean hesitation of the pair, emphasising uncertain points when $\lambda > 0$ (Xu et al. style modification).

Usage

```
ifcmeans(
  x,
  k,
  m = 2,
  lambda = 0,
  maxit = 100,
  tol = 1e-05,
  nstart = 1L,
  seed = NULL
)
```

Arguments

x	object coercible to ifv (n observations).
k	number of clusters.
m	fuzzifier > 1 (default 2).
lambda	hesitation penalty ≥ 0 (default 0 = plain IFCM).
maxit	maximum iterations.
tol	convergence tolerance on membership change.
nstart	number of random starts (best kept).
seed	random seed.

Value

A list with centers (ifv, k rows), membership (n x k matrix), cluster (hard labels), within_ss (sum of weighted squared distances), iter, converged.

References

Xu, Z., Chen, J. & Wu, J. (2008). Clustering algorithm for intuitionistic fuzzy sets. *Information Sciences*, 178(19), 3775-3790.

Examples

```
set.seed(1)
mu <- runif(30, 0.3, 0.7); nu <- runif(30, 0.05, 0.25)
x <- ifv(mu, nu)
r <- ifcmeans(x, k = 2, nstart = 2)
r$cluster; r$centers
```

ifkmeans

*Intuitionistic fuzzy k-means (hard labels, IF centroids)***Description**

Lloyd-style k-means with normalised Euclidean IF distance and IFWA centroid updates.

Usage

```
ifkmeans(x, k, maxit = 50, nstart = 5L, seed = NULL)
```

Arguments

x	object coercible to ifv.
k	number of clusters.
maxit	maximum iterations.
nstart	random starts.
seed	seed.

Value

list with centers (ifv), cluster, within_ss, iter, converged.

Examples

```
set.seed(2)
mu <- runif(20, 0.2, 0.7); nu <- runif(20, 0.0, 0.25)
x <- ifv(mu, nu)
ifkmeans(x, k = 2, nstart = 2)
```

ifv	<i>Create an intuitionistic fuzzy value (IFV)</i>
-----	---

Description

An intuitionistic fuzzy value is a pair (μ, ν) with $\mu \geq 0$, $\nu \geq 0$ and $\mu + \nu \leq 1$. Hesitation is $\pi = 1 - \mu - \nu$.

Usage

```
ifv(mu, nu)
```

Arguments

mu	numeric vector of membership degrees in $[0, 1]$.
nu	numeric vector of non-membership degrees in $[0, 1]$.

Value

An object of class "ifv": a data.frame with columns mu, nu, pi.

Examples

```
ifv(0.6, 0.3)
ifv(c(0.5, 0.7), c(0.3, 0.1))
```

ifwa

*Intuitionistic fuzzy aggregation operators***Description**

Weighted / ordered / hybrid aggregation of IFVs after Xu (2007) and Xu & Yager (2006).

Usage

ifwa(x, w)

ifwg(x, w)

ifowa(x, order_w)

ifowg(x, order_w)

ifha(x, w, order_w)

ifhg(x, w, order_w)

if_einstein_wa(x, w)

if_einstein_wg(x, w)

Arguments

x	object coercible to ifv (vector of values to fuse).
w	numeric weight vector (non-negative, sums to 1; normalised if not).
order_w	positional weights for ordered/hybrid operators.

Details

For IFVs $a_j = (\mu_j, \nu_j)$ with weights w_j :

$$IFWA = \left(1 - \prod (1 - \mu_j)^{w_j}, \prod \nu_j^{w_j} \right)$$

$$IFWG = \left(\prod \mu_j^{w_j}, 1 - \prod (1 - \nu_j)^{w_j} \right)$$

Ordered variants (IFOWA/IFOWG) first reorder a_j by score $S = \mu - \nu$ (descending) and then apply positional weights. Hybrid variants (IFHA/IFHG) weight each a_j by nw_j (via scalar-mul / power) and then apply ordered weighting.

Terms 0^0 arising from zero weights are treated as 1, and 0^0 -type products skip zero-weight factors, so unused inputs do not affect the result.

Value

A single-row ifv.

References

- Xu, Z. (2007). Intuitionistic fuzzy aggregation operators. *IEEE Trans. Fuzzy Systems*, 15(6), 1179-1187.
- Xu, Z. & Yager, R. R. (2006). Some geometric aggregation operators based on intuitionistic fuzzy sets. *Int. J. General Systems*, 35(4), 417-433.

Examples

```
x <- ifv(c(0.6, 0.5, 0.7), c(0.3, 0.2, 0.1))
ifwa(x, c(0.5, 0.3, 0.2))
ifwg(x, c(0.5, 0.3, 0.2))
ifowa(x, c(0.5, 0.3, 0.2))
```

if_add	<i>Intuitionistic fuzzy arithmetic (Xu operations)</i>
--------	--

Description

Addition, multiplication, scalar multiplication and power for IFVs, following Xu (2007): for $a = (\mu_a, \nu_a)$, $b = (\mu_b, \nu_b)$, $\lambda > 0$:

$$a \oplus b = (\mu_a + \mu_b - \mu_a \mu_b, \nu_a \nu_b)$$

$$a \otimes b = (\mu_a \mu_b, \nu_a + \nu_b - \nu_a \nu_b)$$

$$\lambda a = (1 - (1 - \mu_a)^\lambda, \nu_a^\lambda)$$

$$a^\lambda = (\mu_a^\lambda, 1 - (1 - \nu_a)^\lambda)$$

Usage

```
if_add(a, b)

if_mul(a, b)

if_scalar_mul(lambda, a)

if_pow(a, lambda)

if_einstein_add(a, b)

if_einstein_mul(a, b)

if_einstein_scalar_mul(lambda, a)

if_einstein_pow(a, lambda)
```

Arguments

a, b objects coercible to ifv (recycled to common length).

lambda positive numeric scalar or vector.

Details

Einstein variants use Einstein t-norm / t-conorm:

$$a \oplus_{\epsilon} b = \left(\frac{\mu_a + \mu_b}{1 + \mu_a \mu_b}, \frac{\nu_a \nu_b}{1 + (1 - \nu_a)(1 - \nu_b)} \right)$$

$$a \otimes_{\epsilon} b = \left(\frac{\mu_a \mu_b}{1 + (1 - \mu_a)(1 - \mu_b)}, \frac{\nu_a + \nu_b}{1 + \nu_a \nu_b} \right)$$

Value

An ifv object.

References

Xu, Z. (2007). Intuitionistic fuzzy aggregation operators. *IEEE Trans. Fuzzy Systems*, 15(6), 1179-1187.

Examples

```
a <- ifv(0.6, 0.3); b <- ifv(0.5, 0.3)
if_add(a, b); if_mul(a, b); if_scalar_mul(2, a); if_pow(a, 2)
```

if_aggregate_matrix	<i>Aggregate rows of an IF decision matrix</i>
---------------------	--

Description

Aggregate rows of an IF decision matrix

Usage

```
if_aggregate_matrix(mat, w, ncrit, fun = ifwa)
```

Arguments

mat	ifv with one row per alternative-criterion cell, or a list of per-alternative ifv vectors; see examples.
w	criterion weights.
ncrit	number of criteria (columns); rows of mat are nalt blocks of ncrit in row-major order.
fun	aggregation function, e.g. ifwa, ifwg.

Value

ifv vector (one fused value per alternative).

Examples

```
m <- ifv(c(0.6, 0.5, 0.7, 0.4), c(0.2, 0.3, 0.1, 0.4))
# 2 alternatives x 2 criteria stored row-wise:
if_aggregate_matrix(m, w = c(0.6, 0.4), ncrit = 2, fun = ifwa)
```

`if_check`*Validate IFVs*

Description

Validate IFVs

Usage

```
if_check(x, tol = 1e-09)
```

Arguments

<code>x</code>	object coercible to ifv.
<code>tol</code>	numeric tolerance.

Value

invisibly TRUE; errors if invalid.

`if_comp`*Complement of an IFV*

Description

Complement of an IFV

Usage

```
if_comp(x)
```

Arguments

<code>x</code>	object coercible to ifv.
----------------	--------------------------

Value

ifv with mu/nu swapped.

Examples

```
if_comp(ifv(0.6, 0.2))
```

if_compose

*Intuitionistic fuzzy relations: composition and inference***Description**

Max-min / max-product composition of IF relations and the compositional rule of inference (Zadeh, 1973, extended to IFS).

Usage

```
if_compose(R, S, nx, ny, nz, op = c("maxmin", "maxprod"))
```

Arguments

R, S	matrices (or data.frames) of IFVs stored as ifv objects with nrow = nx*ny in row-major order, or numeric matrices of membership values with matching non-membership matrices R_nu, S_nu. Simpler: pass ifv vectors plus dimensions.
nx, ny, nz	dimensions.
op	"maxmin" or "maxprod".

Details

An IF relation on $X \times Y$ is a matrix of IFVs. Composition $R \circ S$ on $X \times Z$ is:

$$(R \circ S)(x, z) = \bigoplus_y (R(x, y) \otimes S(y, z))$$

where $\oplus, \otimes$ are max-min (default) or max-product. Membership composes with t-conorm over t-norm, non-membership with the dual operation (min over max for max-min; product-sum dual for max-product).

Value

An ifv vector of length nx*nz (row-major).

Examples

```
R <- ifv(c(0.6, 0.3, 0.8, 0.2), c(0.2, 0.5, 0.1, 0.6)) # 2x2
S <- ifv(c(0.5, 0.7, 0.4, 0.9), c(0.3, 0.1, 0.4, 0.0)) # 2x2
if_compose(R, S, nx = 2, ny = 2, nz = 2)
```

if_defuzzify	<i>Defuzzify IFVs to crisp scores</i>
--------------	---------------------------------------

Description

Defuzzify IFVs to crisp scores

Usage

```
if_defuzzify(x, method = c("score", "expected", "accuracy", "mu"))
```

Arguments

x	object coercible to ifv.
method	"score" ($\mu - \nu$), "accuracy" ($\mu + \nu$), "expected" ($\mu + \pi/2$), or "mu" (membership only).

Value

numeric vector.

Examples

```
if_defuzzify(ifv(c(0.6, 0.5), c(0.3, 0.2)))
```

if_dist	<i>Distances between intuitionistic fuzzy sets</i>
---------	--

Description

For sets A, B of equal cardinality n with elements $(\mu_i^A, \nu_i^A, \pi_i^A), (\mu_i^B, \nu_i^B, \pi_i^B)$:

Usage

```
if_dist(
  a,
  b,
  method = c("hamming", "euclidean", "hausdorff", "ye", "ye_cosine", "park"),
  w = NULL
)
```

Arguments

a, b	objects coercible to ifv of common length.
method	distance name (partial matching allowed).
w	optional element weights.

Details

- "hamming" (normalised Szmidt-Kacprzyk): $\frac{1}{2n} \sum (|\Delta\mu| + |\Delta\nu| + |\Delta\pi|)$.
- "euclidean" (normalised): $\sqrt{\frac{1}{2n} \sum ((\Delta\mu)^2 + (\Delta\nu)^2 + (\Delta\pi)^2)}$.
- "hausdorff" (normalised Hamming-Hausdorff): $\frac{1}{n} \sum \max(|\Delta\mu|, |\Delta\nu|, |\Delta\pi|)$.
- "ye" (Ye's modified Hamming-type distance): $\sum w_i \frac{|\Delta\mu_i| + |\Delta\nu_i|}{|\Delta\mu_i| + |\Delta\nu_i| + \min(\mu_i^A, \mu_i^B) + \min(\nu_i^A, \nu_i^B) + \min(\pi_i^A, \pi_i^B) + 1}$.
- "ye_cosine": $1 - S_C$ where S_C is Ye's cosine similarity (see [if_sim](#)).
- "park": normalised Hamming including hesitation (same numeric form as "hamming" here).

Weighted versions: pass w (sums to 1) to weight elements: $D_w = \sum w_i d_i$ with per-element d_i .

Value

numeric scalar distance.

References

Szmidt, E. & Kacprzyk, J. (2000). Distances between intuitionistic fuzzy sets. *Fuzzy Sets and Systems*, 114(3), 505-518.

Ye, J. (2012). Multicriteria decision-making method using the Dice similarity measure based on the reduct intuitionistic fuzzy sets of interval-valued intuitionistic fuzzy sets. *Applied Mathematical Modelling*, 36(9), 4466-4472.

Examples

```
a <- ifv(c(0.6, 0.5), c(0.3, 0.2)); b <- ifv(c(0.5, 0.4), c(0.3, 0.3))
if_dist(a, b, "hamming"); if_dist(a, b, "euclidean"); if_dist(a, b, "hausdorff")
```

if_entropy

Entropy of intuitionistic fuzzy sets

Description

Fuzziness / intuitionism of a set $A = \{a_1, \dots, a_n\}$. All measures lie in $[0, 1]$ with larger = more fuzzy.

Usage

```
if_entropy(x, method = c("burillo", "szmidt", "vlachos", "zeng_li"))
```

Arguments

x	object coercible to ifv (the whole set).
method	one of "burillo", "szmidt", "vlachos", "zeng_li" (partial matching allowed).

Details

- "burillo" (Burillo & Bustince, 1996): $E = \frac{1}{n} \sum \pi_i$.
- "szmidt" (Szmidt & Kacprzyk, 2001): $E = \frac{1}{n} \sum \frac{\min(\mu_i, \nu_i) + \pi_i}{\max(\mu_i, \nu_i) + \pi_i}$ (define $0/0 := 1$).
- "vlachos" (Vlachos & Sergiadis, 2007; De Luca-Termini extension): $E = \frac{-1}{n \ln 2} \sum [\mu_i \ln \mu_i + \nu_i \ln \nu_i - (1 - \pi_i) \ln(1 - \pi_i) - \pi_i \ln 2]$, with $0 \ln 0 := 0$.
- "zeng_li" (Zeng & Li, 2006): $E = \frac{1}{n} \sum (1 - |\mu_i - \nu_i|)$.

Value

numeric scalar in $[0, 1]$.

References

- Burillo, P. & Bustince, H. (1996). Entropy on intuitionistic fuzzy sets and on interval-valued fuzzy sets. *Fuzzy Sets and Systems*, 78(3), 305-316.
- Szmidt, E. & Kacprzyk, J. (2001). Entropy for intuitionistic fuzzy sets. *Fuzzy Sets and Systems*, 118(3), 467-477.
- Vlachos, I. K. & Sergiadis, G. D. (2007). Intuitionistic fuzzy information - applications to pattern recognition. *Pattern Recognition Letters*, 28(2), 197-206.
- Zeng, L. & Li, H. (2006). Normalized distance, similarity measure, intuitionistic fuzzy entropy and application. Proc. World Congress on Intelligent Control and Automation, 3857-3860.

Examples

```
x <- ifv(c(0.5, 0.6, 0.4), c(0.3, 0.2, 0.4))
if_entropy(x, "burillo"); if_entropy(x, "szmidt")
if_entropy(x, "vlachos"); if_entropy(x, "zeng_li")
```

if_fuzzify

Fuzzify numeric data into intuitionistic fuzzy values

Description

Converts a numeric vector into IFVs by normalising into $[0, 1]$ to get membership, then deriving non-membership as $\nu = (1 - \mu) \times \nu_ratio$ so hesitation absorbs the remainder. Useful for feeding measurements into the IF toolkit.

Usage

```
if_fuzzify(x, mu_range = range(x, na.rm = TRUE), nu_ratio = 0.6)
```

Arguments

- | | |
|----------|---|
| x | numeric vector. |
| mu_range | range used to scale x into membership (default range(x)). |
| nu_ratio | share of the non-membership part in $(1 - \mu)$ (scalar or vector in $[0, 1]$, default 0.6). |

Value

ifv vector.

Examples

```
if_fuzzify(1:5)
if_fuzzify(c(10, 20, 30), nu_ratio = 0.5)
```

if_infer	<i>Compositional rule of inference for IFVs</i>
----------	---

Description

Given a fact (IF set on X , length n_x) and an IF implication relation on $X \times Y$ ($n_x \times n_y$ values row-major), infer the consequent on Y via max-min composition.

Usage

```
if_infer(fact, relation, nx, ny, op = c("maxmin", "maxprod"))
```

Arguments

fact	ifv of length n_x .
relation	ifv of length $n_x \times n_y$.
n_x, n_y	dimensions.
op	composition operator.

Value

ifv of length n_y .

Examples

```
fact <- ifv(c(0.8, 0.2), c(0.1, 0.7))
rel <- ifv(c(0.6, 0.3, 0.8, 0.2), c(0.2, 0.5, 0.1, 0.6))
if_infer(fact, rel, nx = 2, ny = 2)
```

if_pi	<i>Hesitation degree</i>
-------	--------------------------

Description

Hesitation degree

Usage

```
if_pi(x)
```

Arguments

x an ifv object (or coercible).

Value

numeric vector of $\pi = 1 - \mu - \nu$.

if_rank_df	<i>Rank alternatives from aggregated IFVs</i>
------------	---

Description

Rank alternatives from aggregated IFVs

Usage

```
if_rank_df(x, labels = NULL)
```

Arguments

x object coercible to ifv (one value per alternative).
 labels optional alternative labels.

Value

ranked data.frame.

Examples

```
if_rank_df(ifv(c(0.6, 0.5, 0.7), c(0.3, 0.2, 0.1)))
```

if_rule_block	<i>Mamdani-style IF rule block evaluation</i>
---------------	---

Description

Evaluate nrules rules, each mapping an nx-vector input fact through its own nx*ny relation, then aggregate consequents by element-wise max (mu) / min (nu) union.

Usage

```
if_rule_block(fact, relations, nx, ny, op = c("maxmin", "maxprod"))
```

Arguments

fact	ifv length nx.
relations	list of nrules ifv relations (each length nx*ny).
nx, ny	dimensions.
op	composition operator.

Value

ifv length ny.

if_score	<i>Score, accuracy and ranking of IFVs</i>
----------	--

Description

Score $S = \mu - \nu \in [-1, 1]$, accuracy $H = \mu + \nu \in [0, 1]$ (Chen & Tan, 1994; Hong & Choi, 2000). Ranking is lexicographic: larger score first, ties broken by larger accuracy, then larger hesitation-handling via tie = "pi_min" (smaller hesitation wins).

Usage

```
if_score(x)
if_accuracy(x)
if_hesitation(x)
if_rank(x)
if_order(x)
if_best(x)
```

Arguments

x	object coercible to ifv.
---	--------------------------

Value

if_score()/if_accuracy()/if_hesitation() numeric vectors; if_rank() integer ranks (1 = best); if_order() ordering permutation (best first); if_best() best element(s).

References

Chen, S. M. & Tan, J. M. (1994). Handling multicriteria fuzzy decision-making problems based on vague set theory. *Fuzzy Sets and Systems*, 67(2), 163-172.

Hong, D. H. & Choi, C.-H. (2000). Multicriteria fuzzy decision-making problems based on vague set theory. *Fuzzy Sets and Systems*, 114(1), 103-113.

Examples

```
x <- ifv(c(0.6, 0.5, 0.7), c(0.3, 0.2, 0.1))
if_score(x); if_accuracy(x); if_rank(x); if_order(x)
```

if_sim	<i>Similarity between intuitionistic fuzzy sets</i>
--------	---

Description

Similarities in $[0, 1]$ (larger = more similar), complementing [if_dist](#).

Usage

```
if_sim(
  a,
  b,
  method = c("chen", "hong_kim", "ye_cosine", "ye_weighted", "deng", "liang_shi", "dice"),
  w = NULL
)
```

Arguments

a, b objects coercible to ifv.
 method similarity name (partial matching allowed).
 w optional element weights.

Details

- "chen": $1 - D_H$ (one minus normalised Hamming).
- "hong_kim": $1 - D_E$ (one minus normalised Euclidean).
- "ye_cosine" (Ye, 2011): weighted cosine $S_C = \sum w_i \frac{\mu_A \mu_B + \nu_A \nu_B + \pi_A \pi_B}{\sqrt{\mu_A^2 + \nu_A^2 + \pi_A^2} \sqrt{\mu_B^2 + \nu_B^2 + \pi_B^2}}$.
- "ye_weighted" (Ye, 2010 correlation): $S_W = \frac{C(A,B)}{\max(C(A,A), C(B,B))}$ with $C(A, B) = \sum w_i (\mu_A \mu_B + \nu_A \nu_B + \pi_A \pi_B)$.
- "deng" / "liang_shi": Jaccard-style $S_J = \sum w_i \frac{\mu_A \mu_B + \nu_A \nu_B + \pi_A \pi_B}{(\mu_A^2 + \nu_A^2 + \pi_A^2) + (\mu_B^2 + \nu_B^2 + \pi_B^2) - (\mu_A \mu_B + \nu_A \nu_B + \pi_A \pi_B)}$.
- "dice": Dice-style $S_D = \sum w_i \frac{2(\mu_A \mu_B + \nu_A \nu_B + \pi_A \pi_B)}{(\mu_A^2 + \nu_A^2 + \pi_A^2) + (\mu_B^2 + \nu_B^2 + \pi_B^2)}$.

Value

numeric scalar similarity in $[0, 1]$.

References

- Ye, J. (2011). Cosine similarity measures for intuitionistic fuzzy sets and their applications. *Mathematical and Computer Modelling*, 53(1-2), 91-97.
- Ye, J. (2010). Fuzzy decision-making method based on the weighted correlation coefficient under intuitionistic fuzzy environment. *European J. Operational Research*, 205(1), 202-204.
- Liang, Z. & Shi, P. (2003). Similarity measures on intuitionistic fuzzy sets. *Pattern Recognition Letters*, 24(15), 2687-2693.

Examples

```
a <- ifv(c(0.6, 0.5), c(0.3, 0.2)); b <- ifv(c(0.5, 0.4), c(0.3, 0.3))
if_sim(a, b, "chen"); if_sim(a, b, "ye_cosine"); if_sim(a, b, "dice")
```

if_topsis

*Intuitionistic fuzzy TOPSIS and decision helpers***Description**

IF-TOPSIS after Boran et al. (2009): aggregate each alternative's criterion IFVs with weights w (via `ifwa` by default), then measure separation from the positive ideal $A^+ = (1, 0)$ and negative ideal $A^- = (0, 1)$ using normalised Euclidean IF distance. Closeness $C_i = D_i^- / (D_i^+ + D_i^-)$; larger is better.

Usage

```
if_topsis(mat, nalt, ncrit, w, dist = "euclidean", agg = ifwa)
```

Arguments

<code>mat</code>	ifv of length <code>nalt*ncrit</code> in row-major order (alternative 1's criteria first), or an <code>nalt x ncrit</code> list-matrix.
<code>nalt, ncrit</code>	dimensions (guessed from square if missing is impossible, so both are required).
<code>w</code>	criterion weights (length <code>ncrit</code>).
<code>dist</code>	distance method passed to <code>if_dist</code> (default "euclidean").
<code>agg</code>	aggregation function (default <code>ifwa</code>).

Value

data.frame with columns `alt`, `mu`, `nu`, `score`, `d_plus`, `d_minus`, `closeness`, `rank`, ordered by `rank`.

References

- Boran, F. E., Genc, S., Kurt, M. & Akay, D. (2009). A multi-criteria intuitionistic fuzzy group decision making for supplier selection with TOPSIS method. *Expert Systems with Applications*, 36(8), 11363-11368.

Examples

```
m <- ifv(c(0.6, 0.5, 0.7, 0.4, 0.5, 0.6), c(0.2, 0.3, 0.1, 0.4, 0.3, 0.2))
if_topsis(m, nalt = 3, ncrit = 2, w = c(0.6, 0.4))
```

is_ifv

Test / coerce intuitionistic fuzzy values

Description

Test / coerce intuitionistic fuzzy values

Usage

```
is_ifv(x)
```

```
as_ifv(x, ...)
```

Arguments

x	object to test or coerce.
...	ignored.

Value

is_ifv() logical; as_ifv() an ifv object.

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